

SIMATS ENGINEERING

ASSESSMENT TOOL 3 – ARTIFICIAL INTELLIGENCE (CSA17)

Descriptive Written Test | Total Marks: 20

Name: _____ Register No.: _____

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Question 1

A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

Answer

Introduction: An intelligent agent perceives its environment through sensors and takes actions through actuators to achieve specific objectives. In a smart home, sensors collect information about motion, temperature, light levels, and time, while actuators control lights, thermostats, locks, and alarms.

1. Simple Reflex Agent

A simple reflex agent selects actions using condition–action rules based on the current input. It does not maintain an internal state or use past experiences. For example, if motion is detected in a dark room, it switches on the light. If smoke is detected, it activates an alarm. It is fast and simple, but cannot independently remember past events or learn user preferences.

2. Model-Based Agent

A model-based agent maintains an internal state of the environment and updates it using current sensor readings and previous observations. For example, if a person enters a bedroom and then remains still, the agent may infer that the room is still occupied even when motion is no longer detected. This helps when sensor information is incomplete, although performance depends on the accuracy of the internal model.

3. Goal-Based Agent

A goal-based agent chooses actions according to whether they help achieve a specified goal. For example, if the goal is to maintain a room temperature of 24°C, it may activate the air conditioner when the temperature rises above the desired level. It can compare possible actions, but may not select the most energy-efficient option when several actions achieve the same goal.

4. Utility-Based Agent

A utility-based agent evaluates possible actions using a utility function that represents the desirability of different outcomes. For example, it can balance comfort, electricity consumption, operating cost, and security before adjusting the air conditioner or lighting. It handles competing objectives effectively, but designing a suitable utility function can be complex.

Comparison

Agent type	Smart home application
Simple reflex	Switches on lights when motion is detected.
Model-based	Maintains an estimate of room occupancy.
Goal-based	Maintains the desired room temperature.
Utility-based	Balances comfort, energy efficiency, and security.

Most Effective Approach

A hybrid approach combining model-based, goal-based, and utility-based agents would be most effective. The model-based component maintains an internal representation of the home, the goal-based component works toward desired outcomes, and the utility-based component balances comfort, energy efficiency, and security. A learning mechanism can additionally analyse user behaviour and adapt schedules and preferences. Simple reflex rules can be retained for immediate safety responses, such as activating an alarm when smoke is detected.

Conclusion: A hybrid intelligent agent provides context-aware and efficient smart home automation while balancing comfort, energy consumption, and security.

Question 2

A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Answer

Introduction: Uninformed search algorithms explore possible states without using a heuristic to estimate how close a state is to the goal. In a maze, each location can be represented as a state and each valid movement as an action. The robot explores states until it reaches the exit.

1. Uniform Cost Search (UCS)

Uniform Cost Search expands the frontier state with the lowest accumulated path cost. It is useful when movements have different costs, such as when a hazardous tile costs more than a normal tile.

Working:

- Start at the entrance with path cost zero and add it to a priority queue.
- Remove the state with the lowest total path cost.
- If it is the exit, return the path.
- Otherwise, examine valid neighbours and update their costs when a cheaper route is found.
- Repeat until the exit is found or no reachable states remain.

Advantages: UCS finds a least-cost path when step costs are non-negative under standard assumptions. It is complete when step costs have a positive lower bound. It handles routes with different movement costs.

Disadvantages: It may explore many states and store a large frontier, consuming substantial memory. It can be unnecessary when all moves have equal cost.

2. Iterative Deepening Search (IDS)

IDS repeatedly performs depth-limited depth-first searches, increasing the depth limit one level at a time. It first searches to depth 0, then depth 1, depth 2, and continues until the exit is found.

Advantages: IDS uses relatively little memory, is complete when the branching factor is finite and a finite-depth solution exists, and finds a shallowest solution when all actions have equal cost.

Disadvantages: It repeats exploration of shallow states at each iteration and does not guarantee the least-cost path when movement costs differ.

3. Time and Space Complexity

Feature	UCS	IDS
Selection method	Lowest accumulated path cost	Depth limit increased gradually
Time	Exponential in path cost in the general search-tree bound; commonly $O(b^{(1 + C^*/\epsilon)})$.	$O(b^d)$
Space	Exponential in path cost in the general search-tree bound.	$O(bd)$
Complete?	Yes, with a positive minimum step cost.	Yes, with finite branching and a finite-depth solution.
Optimal?	Yes, under standard non-negative-cost assumptions.	Yes, if every step has equal cost.

Here, b is the branching factor, d is the depth of the shallowest solution, C^* is the optimal solution cost, and ϵ is the minimum positive step cost. These are general search-tree bounds; actual performance depends on the maze and implementation. In a finite graph-search implementation, UCS complexity can also be expressed in terms of vertices and edges.

4. Relationship with Intelligent Agents

A simple reflex agent may follow rules such as turning left whenever possible, but without memory it may revisit locations and struggle with complex mazes. A model-based agent maintains information about explored locations, obstacles, and visited states. A goal-based agent directs its actions toward reaching the exit. A utility-based agent can compare routes by travel cost, energy use, or danger.

5. Most Effective Combination

For a maze where every movement has equal cost and memory is limited, a model-based, goal-based agent using IDS is effective because it finds a shallowest solution with relatively low memory use. If movements have different costs, a model-based, goal-based agent using UCS is preferable when the objective is to find the minimum-cost path.

For a general robot that must consider safety, travel cost, and energy consumption, a model-based, goal-based agent using UCS is a suitable choice when movement costs are known or can be estimated and are non-negative. The internal model helps manage the unknown maze, the goal guides the robot toward the exit, and UCS prioritises the least-cost route. The robot must update its map and replan if it discovers unexpected obstacles.

Conclusion: IDS is suitable when all moves have equal cost and memory efficiency is important. UCS is suitable when minimising total path cost is the priority. Combining the appropriate search strategy with a model-based, goal-based agent improves systematic exploration and decision-making.